Rambod Ghandiparsi

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Google Scholar | LinkedIn | GitHub | Website

RESEARCH PROFILE

Research Summary. My research lies at the intersection of Software Engineering, Requirements Engineering, software evolution, and AI-assisted software engineering. I use large language models, natural language processing, software repository mining, testing, and empirical methods to recover stakeholder intent, improve requirements quality, evaluate whether software changes satisfy goals, and validate human- and AI-generated software. My research direction focuses on evidence-grounded software assurance, reliable software evolution, and trustworthy AI-assisted software development.

Research Interests. Software Engineering; Requirements Engineering; AI-Assisted Software Engineering; Large Language Models; Software Evolution and Maintenance; Software Traceability; Software Change Assurance; Software Testing and Regression Analysis; Empirical Software Engineering; Human-Centered Software Engineering.

EDUCATION

Ph.D., Computer Science - UT San Antonio **Jan 2024 - Expected Dec 2028**

- Research area: AI-Assisted Software Engineering; Software Evolution and Maintenance; Software Change Assurance; Advisor: Dr. Xiaoyin Wang

M.S., Data Analytics - UT San Antonio **Aug 2022 - Dec 2023**

Master of Business Administration - K. N. Toosi University of Technology **Sep 2015 - Aug 2017**

B.S., Electrical Engineering - Tafresh University **Sep 2008 - Jun 2013**

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Doctoral Researcher - Software Evolution, Reliability, and Security Lab, UT San Antonio **Sep 2026 - Present**

Supervisor: Dr. Xiaoyin Wang

- Investigate AI-assisted software evolution and maintenance, software traceability, software change assurance, and software testing using Open-Source Software artifacts.

Doctoral Researcher - Data-Driven Software Engineering Lab, UT San Antonio **Jan 2024 - Aug 2026**

Supervisor: Dr. Mitra Bokaei Hosseini

- Investigate requirements quality, stakeholder goals, traceability, regression bugs, and software evolution using GitHub issues, pull requests, commits, tests, and release artifacts.

Graduate Teaching Assistant II - Department of Computer Science, UT San Antonio **Sep 2025 - Present**

Support undergraduate and graduate courses in data mining, machine learning, software engineering, cryptography, secure software development, systems programming, and web technologies.

- Assist with course administration, student support, grading, and technical feedback across multiple course sections.

Graduate Teaching Assistant I - Department of Computer Science, UT San Antonio **Sep 2024 - Aug 2025**

Supported undergraduate students across multiple sections of Principles of Cybersecurity course, providing instructional assistance, academic guidance, and coursework support.

- Assisted with instructional support, assessment, grading, and responding to student questions on cybersecurity concepts and coursework.

Graduate Research Assistant - Data-Driven Software Engineering Lab, UT San Antonio **Oct 2023 - Aug 2024**

Supervisor: Dr. Mitra Bokaei Hosseini

- Conducted empirical Software Engineering research using LLMs, repository data, designed experiments, developed annotation protocols, performed qualitative, quantitative, and agreement analyses, and created reproducible research artifacts.

Graduate Research Assistant - Carlos Alvarez College of Business, UT San Antonio **Mar 2023 - Sep 2023**

Supervisor: Dr. Juan Manuel Sanchez

- Contributed to empirical research on ESG-based CEO compensation through data preparation, classification of non-financial performance metrics, and analysis of compensation practices and earnings attributes.

Graduate Research Assistant - Human-Centered Natural Language Processing Lab, UT San Antonio **Jan 2023 - May 2023**

Supervisor: Dr. Anthony Rios

- Contributed to text-to-SQL and IoT/network-security research through dataset preparation, data cleaning, model evaluation, error analysis, and manuscript preparation.

PEER-REVIEWED PUBLICATIONS

Conference Papers

1. Pragyan K C, Rambod Ghandiparsi, Thomas Herron, John Heaps, and Mitra Bokaei Hosseini. “Demystifying Feature Requests: Leveraging LLMs to Refine Feature Requests in Open-Source Software.” IEEE International Requirements Engineering Conference (RE), 2025, pp. 104–116. DOI: 10.1109/RE63999.2025.00020.

Workshop Papers

2. Rambod Ghandiparsi, Oluwaseyi Lawal, Pragyan K C, and Mitra Bokaei Hosseini. “Regression Bugs in Post-Release GitHub Issue Reports: A Pilot Study.” CrowdRE at IEEE Requirements Engineering Workshops (REW), 2026, pp. 148–154. DOI: 10.1109/REW72749.2026.00030.

3. Pragyan K C, Rambod Ghandiparsi, Thomas Herron, John Heaps, and Mitra Bokaei Hosseini. “Towards Better Requirements from the Crowd: Developer Engagement with Feature Requests in Open Source Software.” CrowdRE at IEEE REW, 2025, pp. 123–130. DOI: 10.1109/REW66121.2025.00020.

4. Ryan Pavlich, Nima Ebadi, Richard Tarbell, Billy Linares, Adrian Tan, Rachael Humphreys, Jayanta Kumar Das, Rambod Ghandiparsi, et al. “Beyond Text-to-SQL for IoT Defense: A Comprehensive Framework for Querying and Classifying IoT Threats.” Workshop on Trustworthy NLP (TrustNLP), ACL, 2025, pp. 1–12. DOI: 10.18653/v1/2025.trustnlp-main.1.

5. Pragyan K C, Rambod Ghandiparsi, Rocky Slavin, Sepideh Ghanavati, Travis D. Breaux, and Mitra Bokaei Hosseini. “Toward Regulatory Compliance: A Few-Shot Learning Approach to Extract Processing Activities.” ESPRE at IEEE REW, 2024, pp. 241–250.

SELECTED RESEARCH CONTRIBUTIONS

- Regression-bug analysis: First-authored a study of 100 Mastodon issue reports; identified 20 regressions among 65 confirmed defects and found missing test coverage for 90% of the confirmed regressions.
- LLM-assisted requirements refinement: Co-developed methods to identify ambiguity and incompleteness in feature requests and generate clarification questions, evaluated with human annotations and developer interviews.
- Crowd requirements: Studied 50 open-source feature requests and developer clarification practices to understand how intent, feasibility, and project alignment shape requirements discussions.
- Regulatory compliance and IoT security: Coauthored work on few-shot extraction of GDPR processing activities and the IoT-SQL benchmark with 10,985 text-SQL pairs and 239,398 network-traffic records.

TEACHING EXPERIENCE

Graduate Teaching Assistant - Department of Computer Science, UT San Antonio **Aug 2024 - Present**

- Principles of Cybersecurity (CS 3113); Windows Systems Programming (CS 3523); Web Technologies (CS 4413); Secure Software Development and Analysis (CS 4683); Software Engineering (CS 5103); Cryptography (CS 4363); Applied Cryptography (CS 6373); Data Mining (CS 4373); Machine Learning (CS 4253).

RESEARCH ARTIFACTS, DATA, AND SOFTWARE

- IoT-SQL Dataset: A Benchmark for Text-to-SQL and IoT Threat Classification - public benchmark with 10,985 text-SQL pairs and 239,398 network-traffic records. Zenodo DOI: 10.5281/zenodo.15000588; zenodo.org/records/15000588
- RegressionBugClassificationCrowd - public replication package for the CrowdRE 2026 regression-bug study, including data, classification guidelines, figures, and results. Github.com/rambodparsi/RegressionBugClassificationCrowd

RESEARCH PRESENTATIONS

- “Feature Request Apprentice - Managing Software Changes from Feature Requests.” Los Datos Conference, UT San Antonio, October 2024.

PROFESSIONAL SERVICE

Public Relations Officer (Volunteer) - CSGRAD, UT San Antonio **Aug 2025 - Present**

- Computer Science Graduate Student Association: Coordinate communications, organize workshops and professional development events, and promote graduate student engagement.

Executive Vice President (Volunteer) - Fravahar, Tehran, Iran **Jan 2021 - Jun 2022**

- Tehran Zoroastrian Youth Organization: Led organizational operations and coordinated educational, athletic, and community programs to support youth development and engagement.

AWARDS AND HONORS

- 3rd Place, Graduate Research Poster Competition, Los Datos Conference, UT San Antonio, October 2024.
- Merit-Based Scholarships, Carlos Alvarez College of Business, UT San Antonio, June 2022.
- Leonard and Shirley Sterling Endowed Scholarship for Graduate Students, UT San Antonio, February 2022.

PRIOR PROFESSIONAL EXPERIENCE

Marketing and Sales Manager - Parand Designers and Innovators Ltd. **Jan 2017 - Jun 2022**

- Led marketing and competitor analysis and coordinated with product and R&D teams to translate customer feedback into product improvements.

Product Development Manager / R&D Team Lead - Parand Designers and Innovators Ltd. **May 2013 - Dec 2016**

- Led electronics R&D, prototyping, testing, and product iteration involving embedded and automation systems.

R&D Intern - QeshmVoltage Company **Jun 2012 - Aug 2012**

- Worked with Siemens PLCs, ladder logic, industrial automation, testing, and debugging.

TECHNICAL AND RESEARCH EXPERTISE

Methods. Requirements analysis, stakeholder-goal recovery, software traceability, software-change assessment, regression analysis, repository mining, empirical software engineering, structured annotation, inter-rater agreement, qualitative and quantitative analysis, interviews, surveys, LLM prompting and evaluation, and few-shot learning.

Programming and Tools. Python, SQL, pandas, NumPy, scikit-learn, Git/GitHub, Jupyter, C/C++, MATLAB, R.